

AI & DATA SCIENCE CAREER ROADMAP

6-MONTH PLAN

Personalized roadmap for a Student with Python / Data foundations → targeting a Data Scientist / AI Engineer role

CURRENT POSITION

Status: Student

Core skill: Python / Data (foundational)

Stage: Early builder — has technical entry point but lacks applied ML/AI project depth and portfolio proof.

TARGET GOAL

Role: Data Scientist / AI Engineer

Timeline: 6 months

Definition of success: Job-ready portfolio, 3+ ML/AI projects, interview-ready fundamentals, active professional network.

SKILL GAP ANALYSIS

Area	Have	Need	Priority
Programming	Python basics	Pandas, NumPy, OOP, clean code	High
Statistics & Math	Basic exposure	Prob/stats, linear algebra, calculus for ML	High
Machine Learning	None/minimal	Scikit-learn, model eval, feature engineering	Critical
Deep Learning / AI	None	PyTorch, LLM APIs, NLP/CV basics	High
Tools & Deployment	None	Git, SQL, cloud basics, Streamlit/FastAPI	Medium
Portfolio & Brand	None	GitHub, resume, LinkedIn presence	Critical

RECOMMENDED LEARNING PLAN

Phase	Focus	Key Topics
M1–M2	Foundations	Python mastery, SQL, Pandas/NumPy, statistics, EDA
M3–M4	Machine Learning	Scikit-learn, regression/classification, model evaluation, feature engineering
M5	AI / Deep Learning	PyTorch basics, NLP/LLM APIs, prompt & RAG fundamentals
M6	Deployment & Job Prep	Git/GitHub, model deployment, resume, mock interviews, DSA basics

SUGGESTED PROJECTS / NETWORKING STRATEGY

- EDA + Prediction App** — Kaggle dataset, deployed with Streamlit
- ML Pipeline Project** — end-to-end sklearn pipeline + model comparison, on GitHub
- AI/LLM Mini-App** — RAG chatbot or NLP tool using an LLM API
- Capstone** — real-world dataset, deployed API + dashboard + write-up

- Post weekly build-in-public updates on LinkedIn
- Join 2–3 Data Science/AI communities (Discord, Kaggle, Slack)
- Attend 1 meetup/webinar monthly; connect with 5 practitioners/week
- Seek 1–2 informational interviews or mentors by Month 4

MONTHLY MILESTONES

Month	Milestone
M1	Python + SQL + stats refreshed; GitHub & LinkedIn set up
M2	EDA project completed and published
M3	Core ML algorithms learned; project #2 started
M4	ML pipeline project deployed; first networking contacts made
M5	AI/LLM project built; resume tailored to target roles
M6	Capstone shipped; applying to roles + interview prep active

IMMEDIATE NEXT ACTIONS: 1) Set up GitHub + LinkedIn profiles this week 2) Start Python/Pandas refresher today 3) Pick Project #1 dataset by end of week 4) Join 1 AI/DS community 5) Block 1 hr/day for the next 6 months

Generated using Chain-of-Thought Reasoning